

## CAPE FLATTERY, Australia

WMO: 941880

Lat: 14.967S

Lon: 145.311E

Elev: 63

StdP: 14.66

Time Zone: 10.00 (E10)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 66.8       | 68.7 | 54.5                          | 63.3 | 72.0 | 56.3 | 67.7 | 72.8 | 28.7                  | 73.9 | 27.3 | 73.9 | 7.0                   | 100  | 0.588 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 12            | 10.9                      | 90.5            | 78.3 | 89.4 | 78.0 | 88.5 | 77.6 | 80.9                | 87.1 | 80.1 | 86.4 | 79.4 | 85.6 | 12.8                    | 140  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.1                            | 151.4 | 84.5 | 78.3 | 147.2 | 83.9 | 77.6 | 143.9 | 83.4 | 44.6          | 87.4 | 43.8 | 86.7 | 43.1 | 85.9 | 84.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 25.8              | 24.6 | 23.1 | DB                         | 61.0 | 94.9               | 4.5 | 2.7                                                | 57.8 | 96.8       | 55.2 | 98.4       | 52.7 | 99.9       | 49.4 | 101.9 |  |
|                   |      |      | WB                         | 56.7 | 82.4               | 3.3 | 1.2                                                | 54.4 | 83.2       | 52.5 | 83.9       | 50.6 | 84.6       | 48.2 | 85.4  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 79.2   | 82.4  | 82.4  | 81.6  | 80.5  | 78.2  | 75.4  | 74.5  | 75.1  | 77.3  | 79.2  | 81.5  | 83.0 |
|                                                                                  | DBStd       | 3.54   | 1.93  | 1.95  | 1.87  | 1.73  | 1.88  | 2.33  | 2.09  | 2.00  | 1.85  | 2.01  | 1.62  | 1.79 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 10674  | 1003  | 908   | 980   | 916   | 874   | 763   | 758   | 778   | 819   | 907   | 945   | 1023 |
|                                                                                  | CDD65       | 5199   | 538   | 488   | 515   | 466   | 409   | 313   | 293   | 313   | 369   | 442   | 495   | 558  |
|                                                                                  | CDH74       | 44070  | 6001  | 5514  | 5393  | 4355  | 2804  | 1213  | 858   | 1129  | 2124  | 3482  | 4882  | 6315 |
|                                                                                  | CDH80       | 11470  | 2084  | 1934  | 1521  | 837   | 273   | 50    | 16    | 44    | 266   | 750   | 1413  | 2282 |
| Wind                                                                             | WSAvg       | 14.4   | 10.6  | 10.3  | 12.4  | 16.6  | 17.3  | 17.1  | 17.0  | 16.1  | 15.6  | 15.0  | 13.2  | 11.7 |
| Precipitation                                                                    | PrecAvg     | 58.0   | 10.4  | 11.4  | 13.2  | 6.4   | 2.8   | 1.6   | 1.5   | 0.7   | 0.4   | 0.7   | 2.1   | 6.1  |
|                                                                                  | PrecMax     | 103.8  | 26.0  | 22.2  | 25.6  | 21.7  | 8.9   | 5.7   | 3.8   | 2.6   | 1.1   | 2.9   | 16.1  | 25.3 |
|                                                                                  | PrecMin     | 34.0   | 3.3   | 0.9   | 0.6   | 0.2   | 0.0   | 0.2   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 17.7   | 6.3   | 5.0   | 6.8   | 6.0   | 2.6   | 1.5   | 0.9   | 0.6   | 0.3   | 0.7   | 3.8   | 5.5  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 91.2  | 93.3  | 90.4  | 87.2  | 84.9  | 83.2  | 81.6  | 82.5  | 85.0  | 87.5  | 89.9  | 92.1 |
|                                                                                  |             | MCWB   | 79.2  | 78.5  | 79.1  | 76.7  | 75.3  | 74.3  | 71.6  | 71.9  | 73.9  | 75.8  | 77.2  | 78.4 |
|                                                                                  | 2%          | DB     | 89.7  | 90.0  | 88.5  | 86.0  | 83.6  | 81.4  | 80.3  | 81.2  | 83.7  | 86.1  | 88.2  | 90.3 |
|                                                                                  |             | MCWB   | 78.7  | 79.0  | 78.8  | 76.3  | 74.1  | 73.1  | 70.9  | 70.7  | 72.6  | 74.5  | 76.3  | 77.5 |
|                                                                                  | 5%          | DB     | 88.6  | 88.8  | 87.3  | 85.1  | 82.6  | 80.2  | 79.2  | 80.1  | 82.7  | 85.1  | 87.2  | 89.3 |
|                                                                                  |             | MCWB   | 78.2  | 78.6  | 78.1  | 75.6  | 73.5  | 72.3  | 70.5  | 69.9  | 71.6  | 73.7  | 75.6  | 77.0 |
|                                                                                  | 10%         | DB     | 87.5  | 87.6  | 86.2  | 84.1  | 81.6  | 79.0  | 78.0  | 79.0  | 81.6  | 84.1  | 86.2  | 88.3 |
|                                                                                  |             | MCWB   | 77.8  | 78.2  | 77.6  | 75.0  | 73.1  | 71.8  | 70.2  | 69.3  | 71.0  | 72.9  | 74.8  | 76.5 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 81.4  | 81.9  | 81.6  | 79.7  | 78.2  | 76.4  | 75.3  | 75.4  | 76.5  | 77.3  | 79.6  | 81.1 |
|                                                                                  |             | MCDB   | 88.0  | 88.0  | 87.5  | 83.7  | 81.8  | 80.3  | 78.4  | 78.8  | 82.4  | 85.2  | 86.7  | 87.0 |
|                                                                                  | 2%          | WB     | 80.4  | 80.8  | 80.5  | 78.3  | 76.8  | 74.7  | 73.8  | 73.8  | 75.0  | 76.2  | 78.2  | 79.7 |
|                                                                                  |             | MCDB   | 87.1  | 87.1  | 86.5  | 83.0  | 80.6  | 78.9  | 77.2  | 77.9  | 81.0  | 83.6  | 85.1  | 86.8 |
|                                                                                  | 5%          | WB     | 79.7  | 80.0  | 79.5  | 77.6  | 75.7  | 73.8  | 72.8  | 72.5  | 73.7  | 75.4  | 77.2  | 78.9 |
|                                                                                  |             | MCDB   | 86.2  | 86.2  | 85.2  | 82.4  | 80.1  | 78.1  | 76.5  | 77.4  | 80.1  | 82.1  | 84.7  | 86.2 |
|                                                                                  | 10%         | WB     | 79.0  | 79.2  | 78.8  | 76.9  | 74.9  | 73.1  | 71.9  | 71.2  | 72.6  | 74.6  | 76.3  | 78.0 |
|                                                                                  |             | MCDB   | 85.4  | 85.4  | 84.2  | 81.8  | 79.6  | 77.5  | 76.0  | 76.8  | 79.4  | 81.3  | 83.8  | 85.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.2  | 10.2  | 8.8   | 7.4   | 7.0   | 6.8   | 7.1   | 8.4   | 9.5   | 10.1  | 10.4  | 10.9 |
|                                                                                  |             | MCDBR  | 11.7  | 12.5  | 10.5  | 8.2   | 8.0   | 7.9   | 8.0   | 9.2   | 10.1  | 10.8  | 11.2  | 12.2 |
|                                                                                  |             | MCWBR  | 5.0   | 5.2   | 4.7   | 3.9   | 3.8   | 4.0   | 4.0   | 4.4   | 4.2   | 4.5   | 4.2   | 4.5  |
|                                                                                  | 5% WB       | MCDBR  | 10.8  | 11.2  | 9.8   | 7.4   | 7.0   | 7.2   | 6.4   | 8.0   | 9.2   | 10.4  | 10.6  | 11.3 |
|                                                                                  |             | MCWBR  | 5.2   | 5.5   | 5.0   | 3.9   | 3.9   | 4.0   | 3.6   | 4.0   | 4.0   | 4.6   | 4.2   | 4.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.397  | 0.395 | 0.395 | 0.389 | 0.382 | 0.376 | 0.367 | 0.365 | 0.384 | 0.400 | 0.402 | 0.407 |      |
|                                                                                  | taud        | 2.568  | 2.582 | 2.574 | 2.569 | 2.553 | 2.564 | 2.570 | 2.562 | 2.490 | 2.457 | 2.483 | 2.506 |      |
|                                                                                  | Ebn at Noon | 300    | 298   | 292   | 283   | 274   | 271   | 277   | 286   | 291   | 293   | 297   | 297   |      |
|                                                                                  | Edh at Noon | 34     | 34    | 33    | 31    | 30    | 29    | 29    | 31    | 35    | 37    | 37    | 36    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1896   | 1826  | 1696  | 1629  | 1452  | 1312  | 1409  | 1682  | 1985  | 2162  | 2204  | 2090  |      |
|                                                                                  | RadStd      | 166    | 210   | 119   | 150   | 112   | 88    | 91    | 100   | 74    | 108   | 138   | 214   |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page